DO NOW · AO1 REVISION

Solar System, stars, orbits and satellites

Space physics
Paper 2 · Separate Physics only
25 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Name the main objects in our Solar System.**	The Sun, eight planets, dwarf planets, moons, asteroids and comets	☐
2	**What is a galaxy?**	A huge collection of stars held together by gravity.	☐
3	**Give the name of our galaxy.**	The Milky Way	☐
4	**Name a cloud of dust and gas where stars form.**	A nebula	☐
5 ★	**Name the force that pulls a nebula together to form a star.**	Gravity	☐
6	**What is a protostar?**	A hot ball of gas formed as gravity pulls a nebula together, before fusion starts.	☐
7 ★	**Name the process that begins when a protostar becomes hot enough.**	Nuclear fusion	☐
8 ★	**Explain why a main sequence star is stable.**	The inward pull of gravity is balanced by the outward push from fusion energy.	☐
9	**State what a Sun-sized star becomes after its main sequence stage.**	A red giant, then a white dwarf, then a black dwarf	☐
10	**State what a star much bigger than the Sun becomes after its main sequence stage.**	A red super giant, then a supernova, then a neutron star or a black hole	☐
11 ★	**State how elements up to iron are formed.**	By nuclear fusion in stars.	☐
12 ★	**State how elements heavier than iron are formed.**	In supernovae	☐

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	QUESTION	ANSWER	✓
13 ★	**State what keeps one object in orbit around another.**	Gravity	☐
14	**What is an orbit?**	The curved path of one object around another in space.	☐
15	**Name a natural satellite of Earth.**	The Moon	☐
16	**What is an artificial satellite?**	A human-made object in orbit around a planet.	☐
17 ★	**Explain why an object in orbit is accelerating.**	Its direction, and so its velocity, is always changing.	☐
18 ★	**Explain why the velocity of an object in a circular orbit changes but its speed does not.**	Gravity acts at right angles to its motion, changing its direction but not its speed.	☐
19 ★	**State how the speed of a stable orbit depends on its radius.**	The smaller the radius, the greater the speed.	☐
20	**State what the Sun was formed from.**	A nebula (a cloud of dust and gas) pulled together by gravity.	☐
21	**Name the natural satellites of planets.**	Moons	☐
22 ★	**List the stages in the life cycle of a Sun-sized star.**	Nebula, protostar, main sequence star, red giant, white dwarf, black dwarf	☐
23 ★	**List the stages in the life cycle of a star much bigger than the Sun.**	Nebula, protostar, main sequence star, red super giant, supernova, then a neutron star or a black hole	☐
24 ★	**State how elements made in stars are spread through the Universe.**	By supernova explosions.	☐
25	**What do planets orbit? What do moons and artificial satellites orbit?**	Planets orbit the Sun. Moons and artificial satellites orbit planets.	☐

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